

Astrodynamics I: Midterm Exam

April 20, 2016 11:05-12:35PM (90 minutes)

Closed Books/Notes

Use the following formulae without proof if necessary:

$$\bullet \quad \vec{e} = \frac{1}{\mu} \left[\left(v^2 - \frac{\mu}{r} \right) \vec{r} - (\vec{r} \cdot \vec{v}) \vec{v} \right], \quad e = \sqrt{1 + \frac{2Eh^2}{\mu_E^2}}$$

$$\bullet \quad M = \begin{cases} E - e \sin E & , e < 1 \\ B + \frac{1}{3} B^3 & , e = 1 \\ e \sinh H - H & , e > 1 \end{cases}$$

1. (10) If v_a and v_p are the apoapsis and periapsis speeds of an elliptic orbit, respectively, with semimajor axis a and eccentricity e , show that

A. (5) Their arithmetic mean is $\frac{1}{2}(v_a + v_p) = \sqrt{\frac{\mu}{a(1-e^2)}}$.

B. (5) Their geometric mean is $\sqrt{v_a v_p} = \sqrt{\frac{\mu}{a}}$.

2. (30) An Earth-orbiting artificial satellite in elliptical orbit is observed to have an altitude at perigee of 220 km and an altitude at apogee of 360 km. The gravitational parameter of the Earth $\mu_E = 3.986 \times 10^5 \text{ km}^3 / \text{sec}^2$ and the radius of the Earth $R_E = 6,378 \text{ km}$.

- A. (4) What is the orbital specific angular momentum?
- B. (4) What is the orbital specific energy?
- C. (4) What is the orbital eccentricity?
- D. (4) What are the orbital velocities at perigee and at apogee?
- E. (4) What is the orbital period?
- F. (10) At a true anomaly of 45 degrees as measured from perigee, determine the orbital altitude, the magnitude of the orbital velocity vector, and the flight path angle.

(To be Continued)

3. (20) Show that the following physical quantities are conserved for the two-body problem:
- A. (10) Specific Angular Momentum
 - B. (10) Specific Energy
4. (20) Given $\vec{r} = -\frac{1}{\sqrt{2}}\hat{I} + \frac{1}{\sqrt{2}}\hat{J}$ and $\vec{v} = \frac{1}{2}\hat{J}$, determine the orbital elements and sketch the orbit. Use $\mu = 1$.
5. (20) Given a pair of initial position and velocity $(\vec{r}_0, \vec{v}_0)$, describe the process of solving Kepler's problem using the classical orbital elements.